

BIOLOGY MASTER SYLLABUS

CLASS 11 • NCERT / CBSE

Chapter 1: The Living World

- Characteristics of Living Organisms
- Biodiversity
- Taxonomy
- Systematics
- Binomial Nomenclature
- Taxonomic Categories
- Taxonomic Hierarchy

Chapter 2: Biological Classification

- Kingdom Classification
- Five Kingdom System
- Monera
- Protista
- Fungi
- Viruses
- Viroids
- Lichens

Chapter 3: Plant Kingdom

- Algae
- Bryophytes
- Pteridophytes
- Gymnosperms

Chapter 4: Animal Kingdom

- Basis of Classification
- Levels of Organisation
- Symmetry
- Germ Layers
- Body Cavity
- Segmentation
- Notochord
- Non-Chordates
- Chordates

Chapter 5: Morphology of Flowering Plants

- Root
- Stem
- Leaf
- Inflorescence
- Flower
- Fruit
- Seed
- Floral Formula

- Floral Diagram

Chapter 6: Anatomy of Flowering Plants

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|---------------------------|------------------------------|
| • Tissue System | • Internal Structure of Root |
| • Epidermal Tissue System | • Internal Structure of Stem |
| • Ground Tissue System | • Internal Structure of Leaf |
| • Vascular Tissue System | |

Chapter 7: Structural Organisation in Animals

- Frog

Chapter 8: Cell: The Unit of Life

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| • Cell Theory | • Cell Organelles |
| • Prokaryotic Cell | • Nucleus |
| • Eukaryotic Cell | • Cytoskeleton |
| • Cell Envelope | • Cilia |
| • Cell Membrane | • Flagella |
| • Cell Wall | |

Chapter 9: Biomolecules

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|-------------------------|-----------------|
| • Biomolecules | • Proteins |
| • Primary Metabolites | • Lipids |
| • Secondary Metabolites | • Nucleic Acids |
| • Carbohydrates | • Enzymes |

Chapter 10: Cell Cycle and Cell Division

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| • Cell Cycle | • Meiosis |
| • Interphase | • Significance of Cell Division |
| • Mitosis | |

Chapter 11: Photosynthesis in Higher Plants

- Photosynthetic Pigments
- Photosystems
- Light Reaction
- Photophosphorylation
- Carbon Fixation
- C3 Pathway
- C4 Pathway
- Photorespiration
- Factors Affecting Photosynthesis

Chapter 12: Respiration in Plants

- Cellular Respiration
- Glycolysis
- Krebs Cycle
- Electron Transport System
- Oxidative Phosphorylation
- Fermentation
- Respiratory Quotient

Chapter 13: Plant Growth and Development

- Plant Growth
- Growth Phases
- Growth Rate
- Plant Growth Regulators
- Auxins
- Gibberellins
- Cytokinins
- Absciscic Acid
- Ethylene

Chapter 14: Breathing and Exchange of Gases

- Respiratory Organs
- Human Respiratory System
- Mechanism of Breathing
- Pulmonary Ventilation
- Gas Exchange
- Transport of Gases
- Regulation of Respiration
- Respiratory Disorders

Chapter 15: Body Fluids and Circulation

- Blood
- Lymph
- Blood Groups
- Human Heart
- Cardiac Cycle
- Electrocardiography

- Double Circulation

- Circulatory Disorders

Chapter 16: Excretory Products and Their Elimination

- Excretion
- Human Excretory System
- Nephron
- Urine Formation
- Regulation of Kidney Function
- Micturition
- Dialysis
- Excretory Disorders

Chapter 17: Locomotion and Movement

- Types of Movement
- Skeletal System
- Joints
- Muscular System
- Muscle Contraction
- Locomotory Disorders

Chapter 18: Neural Control and Coordination

- Nervous System
- Neuron
- Central Nervous System
- Peripheral Nervous System
- Autonomic Nervous System
- Synapse

Chapter 19: Chemical Coordination and Integration

- Endocrine System
- Hormones
- Hypothalamus
- Pituitary Gland
- Thyroid Gland
- Parathyroid Gland
- Adrenal Gland
- Pancreas
- Pineal Gland
- Gonads
- Hormonal Disorders